

Xylofy AI Internship Week-1 Project

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House Price Prediction using Machine Learning - Project Summary

1. Project Overview:

For my internship's first week's project at Xylofy AI, I worked on a machine learning regression problem to predict real estate prices. The housing market can be highly unpredictable, so the goal of this project was to see if a machine learning algorithm could accurately estimate the selling price of a house based on its specific characteristics.

I used a dataset containing information on 545 houses. For each house, the dataset provided the final selling price (our target variable) along with 13 different features, such as the total area, number of bedrooms and bathrooms, number of stories, furnishing status, and whether the house had extras like air conditioning, parking, or a preferred neighborhood location.

2. My Step-by-Step Process:

To build a reliable prediction model, I followed a complete, end-to-end data science workflow:

Step 1: Data Loading and Inspection:

First, I imported the necessary Python libraries like Pandas, NumPy, Matplotlib, and Seaborn. I loaded the housing dataset into a Pandas DataFrame and looked at the first few rows to understand its structure. I used the `.info()` and `.describe()` functions to check the data types and get a statistical summary. This showed me that we had a good mix of numerical data (like area) and text-based categorical data (like furnishing status).

Step 2: Data Cleaning:

Before feeding data to an algorithm, it needs to be clean. I checked the dataset for missing values (nulls) and duplicate records. Fortunately, this dataset was already in pretty good shape with no missing values, but I made sure to run a `drop-duplicates` function just to guarantee that every single row represented a unique house.

Step 3: Exploratory Data Analysis (EDA):

This was one of the most interesting parts of the project because I got to visualize the data and look for patterns before doing any actual machine learning.

- Univariate Analysis: I plotted histograms to see the distribution of numerical features. I noticed the house prices were "right-skewed," meaning most homes are moderately priced, but a few really expensive luxury homes stretch the graph to the right.
- Bivariate Analysis: I used scatter plots to compare house area against price, which showed a clear upward trend. I also created boxplots to see how categories like the number of bathrooms or having air conditioning affected the median price.
- Multivariate Analysis: I built a correlation heatmap to see how all the numbers interact with each other. This mathematically proved which features were most tightly linked to the final house price.

Step 4: Data Preprocessing

Machine learning algorithms only understand numbers, not text. Since our dataset had categorical columns like "yes/no" for air conditioning and "furnished/unfurnished," I used a technique called One-Hot Encoding (`pd.get_dummies`). I dropped the first column to avoid the dummy variable trap, which basically means preventing redundant data that could confuse the model.

Step 5: Train-Test Split:

I separated the "price" column (what we want to predict) from the rest of the features. Then, I split the dataset into two parts: 80% of the data was used for "training" to teach the model the patterns, and the remaining 20% was kept hidden as "testing" data to grade the model's accuracy later.

Step 6: Model Building and Evaluation:

I decided to test out two different types of machine learning models to see which one was better for this problem:

1. Linear Regression: A basic but powerful algorithm that tries to draw a straight line through the data to predict prices.
2. Random Forest Regressor: A more complex, advanced algorithm that builds lots of decision trees and averages their guesses together.

I trained both models on the 80% training data and then asked them to predict the prices for the 20% testing data. To grade them, I calculated the Mean Absolute Error (MAE), Root Mean

Squared Error (RMSE), and the R^2 Score (which tells us how much of the price variation the model actually understands).

The results are:

	Model	MAE	RMSE	R^2 Score	Accuracy (%)
0	Linear Regression	9.700434e+05	1.324507e+06	0.652924	65.292426
1	Random Forest	1.021546e+06	1.400566e+06	0.611919	61.191853

3. Observations from the Data

Based on exploring the data and evaluating the models, here are the main things I noticed about the housing dataset:

- House Size is the Biggest Factor: The total area of the property is the strongest indicator of price. Bigger houses naturally cost a lot more.
- Bathrooms Over Bedrooms: Surprisingly, having more bathrooms increases the house price more than adding extra bedrooms. Also, having dedicated parking makes a big difference in value.
- Air Conditioning and Location: Houses with AC and those located in popular or preferred neighborhoods are noticeably more expensive than standard homes.
- Furnished vs Unfurnished: Fully furnished houses do sell for more money, but it's not as big of a deal as foundational things like the house size or location.
- Simple Models Worked Better: I expected the more complex Random Forest model to win, but Linear Regression actually did better! This tells us that the relationship between house features and prices is mostly straightforward and linear. Random Forest probably overfitted because our dataset is pretty small (only 545 rows).
- Hard to Predict Luxury Homes: The "Actual vs Predicted" scatter plot shows that the model is good at guessing the price of normal homes, but it usually guesses too low for really expensive ones. This makes sense because luxury homes probably have special features or luxury designs that aren't recorded in our basic dataset.

4. Final Insights

From this project, I learned exactly what makes a house expensive. The data clearly shows that buyers are mostly paying for more physical space and basic comforts like extra bathrooms and AC.

If I were advising a real estate agency based on this data, my main suggestion would be to highlight the floor area, bathroom count, and air conditioning when marketing a home, since these justify a higher asking price. Also, while machine learning is great for estimating standard house prices quickly, human experts are still needed to price luxury or unique homes properly, since algorithms struggle with unrecorded premium features.

5. Conclusion

This project was a great way to experience a complete machine learning workflow from start to finish. I started by cleaning and visualizing the dataset to understand the housing market trends, and then I prepared the data using one-hot encoding so the algorithms could read it.

By comparing Linear Regression and Random Forest, I learned an important lesson: more complex machine learning models aren't always better. Because the dataset was relatively small and had mostly straight-forward relationships, the simpler Linear Regression model actually gave us more accurate predictions (an R^2 score of about 65%) and smaller error margins.

Overall, this project showed me how helpful machine learning can be in the real world. By using predictive models, buyers, sellers, and agents can get a fast, data-backed estimate of a property's value instead of just guessing based on gut feelings.

-----**THANK YOU**-----